(1) If $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$ then show that

(i) $A^2=\begin{bmatrix}\cos 2\alpha & \sin 2\alpha\\ -\sin 2\alpha & \cos 2\alpha\end{bmatrix}$ (ii) $A^3=\begin{bmatrix}\cos 3\alpha & \sin 3\alpha\\ -\sin 3\alpha & \cos 3\alpha\end{bmatrix}$

Solution : (i) Given , $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$

$$\therefore A^2=A.A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}\times\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$$

$$\therefore A^2=\begin{bmatrix}\cos\alpha\cos\alpha+\sin\alpha(-\sin\alpha) & \cos\alpha\sin\alpha+\sin\alpha\cos\alpha\\ (-\sin\alpha)\cos\alpha+\cos\alpha(-\sin\alpha) & (-\sin\alpha)\sin\alpha+\cos\alpha\cos\alpha\end{bmatrix}$$

$$\therefore A^2=\begin{bmatrix}\cos^2\alpha-\sin^2\alpha & 2\sin\alpha\cos\alpha\\ -2\sin\alpha\cos\alpha & \cos^2\alpha-\sin^2\alpha\end{bmatrix}$$

$$\therefore A^2=\begin{bmatrix}\cos 2\alpha & \sin 2\alpha\\ -\sin 2\alpha & \cos 2\alpha\end{bmatrix}\text{- (1)}$$

(ii) We know that $A^3=A^2.A$

$\therefore A^3=\begin{bmatrix}\cos 2\alpha & \sin 2\alpha\\ -\sin 2\alpha & \cos 2\alpha\end{bmatrix}\cdot\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$ from equation (1) we get.

$$\therefore A^3=\begin{bmatrix}\cos 2\alpha\cos\alpha-\sin 2\alpha\sin\alpha & \cos 2\alpha\sin\alpha+\sin 2\alpha\cos\alpha\\ -\sin 2\alpha\cos\alpha-\cos 2\alpha\sin\alpha & -\sin 2\alpha\sin\alpha+\cos 2\alpha\cos\alpha\end{bmatrix}=\text{.....}$$

(2) Construct a 3×3 matrix $A=[a_{ij}]$ whose elements are given by :

(i) $a_{ij}=i\times j$ (ii) $a_{ij}=(i+j)^2$

(iii) $a_{ij}=\begin{cases}0, if & i\neq j\\ 1, if & i=j\end{cases}$ (iv) $a_{ij}=\begin{cases}0, if & i=j\\ 1, if & i>j\\ -1, if & i<j\end{cases}$

Solution : (i) $a_{ij}=i\times j$, $i=1,2,3. j=1,2,3.$

$a_{11}=1\times 1=1$, $a_{12}=1\times 2=2$, $a_{13}=1\times 3=3$

$a_{21}=2\times 1=2$, $a_{22}=2\times 2=4$, $a_{23}=2\times 3=6$

$a_{31}=3\times 1=3$, $a_{32}=3\times 2=6$, $a_{33}=3\times 3=9$

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Thus , $A=\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 6\\ 3 & 6 & 9\end{bmatrix}$

(ii) $a_{ij}=(i+j)^2,\ \ i=1,2,3.\ j=1,2,3.$

$a_{11}=(1+1)^2=2^2=4$ $\quad a_{12}=(1+2)^2=3^2=9$ $\quad a_{13}=(1+3)^2=4^2=16$

$a_{21}=(2+1)^2=3^2=9$ $\quad a_{22}=(2+2)^2=4^2=16$ $\quad a_{23}=(2+3)^2=25$

$a_{31}=(3+1)^2=16$ $\quad a_{32}=(3+2)^2=25$ $\quad a_{33}=(3+3)^2=36$

$$\therefore\quad A=\begin{bmatrix}4 & 9 & 16\\ 9 & 16 & 25\\ 16 & 25 & 36\end{bmatrix}$$

(iii) $a_{ij}=\begin{cases}0, if & i\neq j\\ 1, if & i=j\end{cases}$

$a_{11}=1$ $\quad a_{21}=0$ $\quad a_{31}=0$

$a_{12}=0$ $\quad a_{22}=1$ $\quad a_{32}=0$

$a_{13}=0$ $\quad a_{23}=0$ $\quad a_{33}=1$

$$A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$$

$$A=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

(iv) $a_{ij}=\begin{cases}0, if & i=j\\ 1, if & i>j\\ -1, if & i<j\end{cases}$ for $i=1,2,3.\quad j=1,2,3.$

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$a_{11} = 0$ $\quad a_{21} = 1$ $\quad a_{31} = 1$

$a_{12} = -1$ $\quad a_{22} = 0$ $\quad a_{32} = 1$

$a_{13} = -1$ $\quad a_{23} = -1$ $\quad a_{33} = 0$

$$\therefore A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

(3) Using properties of determinants show that $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$

Solution : Consider $L.H.S. = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$

By using $R_2 - R_1$ and $R_3 - R_1$

$$= \begin{vmatrix} 1 & a & bc \\ 0 & b-a & ca-bc \\ 0 & c-a & ab-bc \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & bc \\ 0 & -(a-b) & c(a-b) \\ 0 & c-a & -b(c-a) \end{vmatrix}$$

By taking $(a-b)$ common from R_2 and $(c-a)$ common from R_3 we get

$$L.H.S. = (a-b)(c-a)\begin{vmatrix} 1 & a & bc \\ 0 & -1 & c \\ 0 & 1 & -b \end{vmatrix}$$

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By using $R_3 + R_2$

$$L.H.S. = (a-b)(c-a)\begin{vmatrix} 1 & a & bc \\ 0 & -1 & c \\ 0 & 0 & c-b \end{vmatrix}$$

$$L.H.S. = (a-b)(c-a)\left[1((-1)(c-b)-0) - a(0-0) + bc(0-0)\right]$$

$$L.H.S. = (a-b)(c-a)[\ldots b-c \ldots]$$

$$L.H.S. = (a-b)(b-c)(c-a)$$

(4) Using properties of determinants show that

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc + ab + bc + ca$$

Solution :

$$LHS = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$$

$R_1 \to R_1 - (1+a)R_3$, $R_2 \to R_2 - R_3$

$$LHS = \begin{vmatrix} 0 & -a & 1-(1+c)(1+a) \\ 0 & b & -c \\ 1 & 1 & 1+c \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -a & -c-a-ac \\ 0 & b & -c \\ 1 & 1 & 1+c \end{vmatrix}$$

$$= 0 + 0 + 1\begin{vmatrix} -a & -c-a-ac \\ b & -c \end{vmatrix}$$ Expand

$$= ac - b(-c-a-ac)$$

$$= ac + bc + ab + abc$$

$$= abc + ab + bc + ca$$

$$= RHS$$

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(5) If $x \neq y \neq z$ and $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$ then show that $1+xyz=0$.

Solution : Given

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$$

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$C_2 \leftrightarrow C_3$ in 1st determinant

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} (1+xyz) = 0 \quad \text{--(1)}$$

Now

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$R_1 \to R_1 - R_2,\ R_2 \to R_2 - R_3$

$$\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$$

Taking $(x-y)(y-z)$ common from R_1 and R_2 respectively

$$= (x-y)(y-z) \begin{vmatrix} 0 & 1 & x+y \\ 0 & 1 & x+z \\ 1 & z & z^2 \end{vmatrix}$$

$R_2 \to R_2 - R_1$

$$= (x-y)(y-z) \begin{vmatrix} 0 & 1 & x+y \\ 0 & 0 & z-x \\ 1 & z & z^2 \end{vmatrix}$$

From (1) and (2)

$$(x-y)(y-z)(z-x) \begin{vmatrix} 0 & 1 & x+y \\ 0 & 0 & 1 \\ 1 & z & z^2 \end{vmatrix} (1+xyz) = 0$$

$$(x-y)(y-z)(z-x)[1(0-1)-0](1+xyz) = 0$$

$$(x-y)(y-z)(z-x)(1+xyz) = 0$$

But $x \neq y \neq z$

$\therefore (x-y)(y-z)(z-x) \neq 0$

$\therefore 1+xyz = 0$

Hence proved

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(6) Using properties of determinants show that

$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$$

Solution :

$$LHS = \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$R_1 \to R_1 + R_2 + R_3$

$$LHS = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$= (a+b+c) \begin{vmatrix} 0 & 0 & 1 \\ a+b+c & -(a+b+c) & 2b \\ 0 & a+b+c & c-a-b \end{vmatrix} \quad \begin{matrix} C_1 \to C_1 - C_2 \\ C_2 \to C_2 - C_3 \end{matrix}$$

Taking $(a+b+c)$ comman from C_1 and C_2

$$= (a+b+c)^3 \begin{vmatrix} 0 & 0 & 1 \\ 1 & -1 & 2b \\ 0 & 1 & c-a-b \end{vmatrix}$$

$= (a+b+c)^3 [1(1)]$

$= (a+b+c)^3$

$= RHS$

$\therefore LHS = RHS$

Sign of Teacher :

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